

Session 6: CPSA Goals

Slides prepared by UMBC Protocol Analysis Lab

Last edited July 28, 2026

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Common Protocol Security Goals

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Common Protocol Security Goals

- ▶ **Confidentiality:** No unauthorized party is able to access the information sent. For example, an eavesdropping adversary does not learn where Alice and Bob wish to meet.
- ▶ **Agreement:** Both parties agree on some set of data values in the protocol. For example, Alice and Bob agree on a meeting location.

Other common security goals

1. Privacy
2. Anonymity
3. Perfect Forward Secrecy
4. Non-Repudiation

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Confidentiality and Agreement (on the secret) needed.

CPSA Goals

- ▶ Protocol goals as mathematical statements
- ▶ Find strands that satisfy (or don't satisfy) some set of properties
- ▶ May reduce time taken to terminate

Some more tips for reading specifications

- ▶ FIND THE SECURITY GOALS
 - ▶ Are additional SGs needed?
- ▶ Modularize the spec into blocks of messages
 - ▶ “In the first set of exchanges”
 - ▶ “[party1] preshares [data] with [party2]”
- ▶ Ignore nonsense
 - ▶ “[protocol] does not consider [niche usecase] because. . .”
 - ▶ “We assume clients have access to an ALU” (structural vs. implementation)
 - ▶ “Our conformance program handles...”

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- ▶ Let's try modeling it!

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Survey

Fill in details about the survey here!